

Muyu He

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Education

M.S. Computer Science	<i>University of Pennsylvania</i>	Sept 2023 - May 2025
B.A. Philosophy	<i>University of California, Los Angeles</i>	Sept 2018 - Dec 2021

Publications

- **Do Value Vectors in Deep Layers Need Context from the Residual Stream?**
Muyu He^{*}, Yuchen Liu^{*}, Qingya Huang, Li Zhang *Equal contribution
(Under review) Empirical Methods in Natural Language Processing (**EMNLP 2026**)
 - Challenged the standard attention mechanism of computing value vectors from the residual stream, discovering that deep layers benefit from a context-free value vector that represents original token information, and that models decouple the optimal input for keys vs. values
 - Proposed a new attention variant that directly learns a value vector table for each of the last third of layers, eliminating the V cache entirely and surpassing standard attention on both validation loss and average scores across 21 benchmarks on two model sizes (135M and 780M)
- **High-fidelity Simulations of Human Traits for Testing Agents**  8
Muyu He, Anand Kumar, Meghana Arakkal Rajeev, Soumyadeep Bakshi, James Zou, Nazneen Rajani
(Accepted by) Association for Computational Linguistics (**ACL Oral 2026**)
 - Improved a training-free technique for fine-grained, arbitrary control over LLM personality by extracting per-layer activations that correspond to human traits (impatience, confusion, etc.)
 - Developed a robust agentic evaluation framework using activation-controlled human-simulator models, demonstrating effectiveness over LoRA, SFT, and prompting on 1K+ conversations scored by human annotators
- **Evaluating Deductive Reasoning via Detective Games**  5
Yuan Yuan^{*}, **Muyu He**^{*}, Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang *Equal contribution
(Accepted by) Empirical Methods in Natural Language Processing (**EMNLP 2025**)
- **Commonsense-T2I Challenge: Can Text-to-Image Generation Models Understand Commonsense?**  45
Xingyu Fu^{*}, **Muyu He**^{*}, Yujie Lu, William Yang Wang, Dan Roth *Equal contribution
(Accepted by) Conference On Language Modeling 2024 (**COLM 2024**)

Work Experience

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| Research Scientist
<i>Zyphra</i> | June 2026 - Current
<i>San Francisco, US</i> |
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- Conducted foundational research on RL algorithms, inference techniques, and hybrid architectures
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| Research Scientist
<i>Collinear.ai</i> | July 2025 - June 2026
<i>Sunnyvale, US</i> |
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- **Spider** (popular on-policy distillation framework)  105
 - Implemented cross-tokenizer alignment to enable on-policy distillation on any teacher-student model pairs, resolving chat-template issues in existing algorithms by HuggingFace
 - Enabled a token-in-token-out protocol for multi-turn rollouts with robust bug patches for standing tokenizers (e.g. Qwen3), extending on-policy distillation to agentic tool use without mismatch issues
 - **Computer-use RL environments** (provider for 3 frontier labs on CUA capabilities)
 - Developed a robust automated environment generation pipeline spanning task generation, verifier generation, data seeding, and Docker runtime setup across 11 real-world apps (e.g. Slack, Asana, Figma, GitHub)
 - Discovered systematic insights on fundamental failure modes in CUA of frontier models (Claude, GPT), enabling the design of RL-efficient tasks in the environment with 33–66% Pass@1

Open-source Projects

RiddleHe/nanochat ★ 52

- Built a hackable pretraining stack from karpathy/nanochat that supports easy definition of architecture and rigorous FLOP-controlled ablations
- Shipped a mirrored RL stack for RL algorithm research that allows clean objective definition with production-grade vLLM inference serving

RiddleHe/llm-interp ★ 91

- Authored a series of completely reproducible interpretability scripts to conduct model circuit research
- Produced findings on attention sinks (the existence of massive activations in 1-2 dimensions for Qwen3 models only on the first tokens) and an open-source reproduction of Thinking Machines' finding on LLM decode indeterminism

RiddleHe/gpt-oss-alignment ★ 125

- Jailbroke the RL-aligned GPT-OSS-20B model via simple chat-template removal, revealing the fragility of post-training alignment that relies on priors of the assistant chat format
- Provided a popular technique to arbitrarily manipulate responses from GPT-OSS-20B by steering activations corresponding to traits, and implemented an end-to-end framework for training its sparse autoencoders